GEK-36466C

DRIVE SYSTEM INSTRUCTIONS

SPEED CHANNEL
With Speed Amplifier Board
Cat. No. 3S7700PB100
For
TURBINE SUPERVISORY INSTRUMENT SYSTEM

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

(GEK-36466C)

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The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, USA.

INTRODUCTION

These instructions cover the speed measurement channel which is part of a turbine supervisory instrument system. For general instructions covering the supervisory instrument system as a whole, refer to the title and contents page of the applicable turbine supervisory instrument system instructions.

The equipment described in this publication is ordinarily used in conjunction with a toothed wheel and a magnetic sensor, both of which are supplied as part of the turbine. The frequency of the sensor output voltage is proportional to the turbine speed. The number of teeth on the wheel is such as to provide a sensor output frequency at synchronous turbine speed (See Table 1).

The speed amplifier board, which is located in a module within the supervisory instrument cabinet, is of the frequency-actuated type and is not affected by the magnitude of the output voltage from the sensor. Most turbine supervisory instrument systems include a speed recorder which works from the output of the speed amplifier board. In some systems, speed is recorded continuously. In others, speed is recorded while the generator circuit breaker is open, switching to record control-valve position while the breaker is closed. The same recorder may also record eccentricity while the turbine is on turning gear.

The speed amplifier board provides an output for feeding a speed input voltage to a computer, and also an output from an optional indicating instrument. Some boards (see Specifications below) also include a cutoff relay for disabling the eccentricity channel when the turbine is turning faster than 30% to 45% of full scale.

SPECIFICATIONS

Catalog Number, Speed Amplifier Board

3S7700PB100A1 - With eccentricity cutoff relay
3S7700PB100B1 - Without eccentricity cutoff relay
3S7700PB100C1 - Without eccentricity cutoff relay

Scale Rating

0-5000 rpm for 3600 or 3000 rpm turbines
0-2500 rpm for 1800 or 1500 rpm turbines
0-8000 rpm for feed pump turbines

Calibration

The speed-amplifier board was tested and correctly adjusted in the factory prior to shipment of the turbine supervisory instrument system. Ordinarily little or no readjustment should be necessary in the field. The calibration adjustments were performed on the following basis:

Table 1

Model No.	Rated Turbine Speed	Sensor Output Frequency (Hz) At Rated Turbine Speed	At Full Scale
A1, B1	3600	4800	6667
A1, B1	3000	4800	8000
A1, B1	1800	4800	6667
A1, B1	1500	4800	8000
A1, B1	FEED PUMP TURBINE	--	8000
C1	3600	3600	5000
C1	3000	3600	6000
C1	1800	3600	5000
C1	1500	3600	6000

Outputs

The following outputs are provided:

A. One to 5 milliamperes (ma) for operating an indicating instrument.

B. Eight to 40 millivolts (mv) for a digital computer and/or data-logging equipment.

C. Two-tenths to 1.0 milliampere (ma) for recorder.

D. Two to 10 volts (v) with output limited by a 2000 ohm $\pm$ 0.1 percent series resistor.

Any one or more of the outputs A, B, C and D may be used or not used as desired without affecting the other outputs. If one or more of these outputs are not to be used, make no connections to the corresponding cabinet output terminals.

An elevated zero is employed, i.e., an output of 1 ma, 8 mv, 0.2 ma or 2 v corresponds to zero rpm.

If an indicating instrument is used, it is recommended that it be rated 0 to 5 ma, with 0 rpm marked at the 1 ma point, and full-scale speed at 5 ma, with 0 rpm marked at the 1 ma point, and full-scale speed at 5 ma. If two or more indicators are to be used, connect them in series. The resistance of the milliammeter (or the total of all connected in series) must not exceed 2000 ohms. Neither side of the 1 to 5 ma output is at ground potential, and therefore, neither side of the connected load (milliammeters) may be grounded.

The 0.2 to 1.0 ma recorder output is frequently used with a recorder having a 50 ohm input resistor, so that a 10 to 50 mv voltage (drop across the 50 ohm resistor) is provided as the recorder input. However, a recorder input resistor other than 50 ohms may be used with some recorders. The negative side of the 0.2 to 1.0 ma output is grounded within the amplifier, and neither side may be grounded at the recorder.

The source impedance of the 8 to 40 mv output is 40 ohms. It is recommended that the input impedance of the connected load (computer input impedance) be at least 5000 ohms. The negative side of this output is grounded within the amplifier, and neither side may be grounded at the computer.

The maximum current which is allowed to be drawn from the special 2 to 10 v output is 2 ma at full scale output voltage.

Response Time

Response time is approximately 0.15 second for 63 percent response. This does not include the response time of the recorder or indicator.

Output Noise (8 to 40 mv Output)

The peak-to-peak output ripple is less than 0.3% of the full-scale millivolt span at synchronous turbine speed. It is less than 1% of the full-scale millivolt span at 1% of synchronous turbine speed.

Accuracy

Excluding the error of the recorder or indicating instrument, accuracy is $\pm$ 0.5 percent of full scale. Including the recorder or indicating instrument, it is $\pm$ 1.5 percent of full scale.

RECEIVING, HANDLING AND STORAGE

The circuit board is shipped in place in the module within the cabinet. Avoid rough handling of the boards. Avoid touching the printed circuit side of the boards with the fingers, as this may cause fingerprint marks under humid conditions. Handle the circuit boards by the edges. If the board should be removed from its module, be certain to return it to exactly the same position from which it was removed. For the correct position of the boards, see the final assembly drawing which is included in the system instructions.

Connections

Refer to the system interconnection diagram for connections between the cabinet and the sensor, recorder, computer and indicator. Make connections to the sensor and to the computer input with shielded cable, and ground the shields at one end only (the cabinet end) as shown. The sensor should be connected using a cable consisting of a twisted pair of No. 18 AWG conductors within a shield, with an insulating cover over the shield. For other wiring, conventional power-station wiring practice will be satisfactory, using No. 18 AWG insulated wire or larger.

Circuit Board

Check to see that the speed-amplifier circuit board is pushed all the way in as far as it will go to be sure it is fully mated in its receptacle. Note that the circuit board has two trimmer potentiometers. The upper of these, R662, is a zero adjustment, which was properly set at the factory and will ordinarily require little or no readjustment in the field. The lower one, R664, is a span adjustment which was accurately set at the factory and should require little (if any) readjustment in the field. See the Preliminary Test and Adjustments section of these instructions for detailed procedure.

PRELIMINARY TEST AND ADJUSTMENTS

NOTE

On turbine supervisory instrument systems having speed, valve position and eccentricity recorded on a single recorder, if the recorder is not already in the speed-recording mode, it will be necessary to force it into that mode to perform the following tests. If the turbine is on turning gear, temporarily connect the two turning-gear switch terminals (in the cabinet) together with a jumper. If the turbine is running at synchronous speed and the generator circuit breaker is closed, temporarily open the circuit-breaker-contact terminals in the cabinet.

Checking and Adjusting Zero

This test is preferably done with the turbine not running. If it should be necessary to perform the test while the turbine is running, temporarily disconnect the speed sensor from its terminals in the cabinet, and connect these two terminals together with a jumper to assure that there will be no input.

1. Turn ON the power switch and turn ON the chart switch in the speed recorder.

2. Set RANGE SELECTOR on test panel to 5 VDC. Connect the red METER INPUT jack to the blue test jack on the front edge of the speed amplifier board, and connect the black METER INPUT jack to the black test jack on the same board.

3. Turn ON the power switch on the power-supply panel in the cabinet.

4. The meter on the test panel should read 2.00 ± 0.05 V, and the recorder should read 0 rpm. If it does not, adjust the ZERO ADJUST, R662, on the speed amplifier circuit board, for a recorder reading of 0 rpm, within ±0.5 percent of full scale. Note that R662 was properly adjusted at the factory. Ordinarily little or no readjustment should be required in the field.

5. If a speed indicator is connected to the system, it should read 1.00 ma (0 rpm).

6. Reconnect sensor leads if they were disconnected prior to Step 1 above.

Span Adjustment (Turbine Running At Synchronous Speed)

1. Proceed as in Steps 1, 2, and 3 under Checking and Adjusting Zero, except setting RANGE SELECTOR switch to 15 VDC.

2. The following adjustment should be made the first time the turbine is running at synchronous speed. (See note immediately following the Preliminary Tests and Adjustment heading above.)

3. The recorder should read 3600 rpm in the case of a 3600 rpm turbine (or should read synchronous speed for 3000, 1800 or 1500 rpm turbines). The meter on the test panel may be used as a qualitative indication to assure that the circuit is working, and should read approximately 7.8 V for 3600 and 1800 rpm turbines or 6.8 V for 3000 and 1500 rpm turbines.

4. If the recorder does not read synchronous speed, adjust the SPAN ADJUSTMENT, R664, on the speed amplifier board, until the recorder reads synchronous speed within ± 0.5 percent of full scale. Note that R664 was properly adjusted at the factory. Ordinarily, little or no readjustment should be required in the field.

5. If the indicator is connected to the system, it should not read synchronous speed.

PRINCIPLES OF OPERATION

The sensor on the turbine provides an AC voltage whose frequency varies in proportion to the turbine speed. This voltage is applied to terminals A and C of the speed amplifier board (see Figure 1). If the incoming sensor voltage is greater than 2.5 V peak to peak, the peaks are clipped down to approximately this value by CR651, CR652 and CR654. Components R653, C655, R654 & C656 remove any unwanted high-frequency noise. The resulting signal is then applied to the frequency-to-voltage converter, A651, which provides a DC output voltage proportional to the frequency of the input.

The DC voltage from the frequency-to-voltage converter is amplified by A652 and Q651. A zero adjustment is provided by trimmer potentiometer R662, while trimmer potentiometer R664 is a feedback adjustment controlling the span. Transistor Q654 provides the 1 to 5 ma output for an indicating instrument, if one is used.

On speed amplifier boards furnished with a relay K651, the relay is energized when the positive DC output voltage from the emitter of Q651 becomes large enough to cause the current through R668 to exceed the current through R669, thus turning on transistors Q652 and Q653, in turn energizing relay K651. The relay is energized when the output voltage at the blue test jack exceeds 4.5 to 5.4 V. On speed amplifier boards having no relay, Q652, Q653 and their associated components are omitted.

TROUBLESHOOTING

The following are some of the most common troubles which may be encountered, their causes and remedies. The steps mentioned refer to the Preliminary Tests and Adjustments section of this publication.

1. TROUBLE: In Step 4 of Checking and Adjusting Zero, the meter reads 0 volts and the recorder reads below zero rpm (against the stop).

   Probable Cause: Speed amplifier board not in place in the correct position, or not plugged all the way into its receptacle.

   Remedy: Refer to the final assembly drawing, furnished with these instructions, for the correct position of board. Push the board in as far as it will go.

2. TROUBLE: In Step 4 of Checking and Adjusting Zero, the meter reads 2.0 V but the recorder does not read zero rpm, and the recorder reading is not affected when the ZERO ADJ, R662, is turned.

   Probable Cause: Recorder may not be in the speed recording mode but rather may be recording eccentricity or valve position.

   Remedy: See note at the beginning of the Preliminary Test and Adjustment section.

3. TROUBLE: In Step 4 of Checking and Adjusting Zero, the meter reads 2.0 V but the recorder reads below zero (against the stop), and the recorder reading is not affected when the ZERO ADJ, R662, is turned.

   Probable Cause: Wiring between supervisory instrument cabinet and recorder input is open or grounded.

   Remedy: Check and correct wiring to recorder.

4. TROUBLE: In Step 4 of Checking and Adjusting Zero, the 2.0 V reading cannot be obtained by turning the ZERO ADJ, R662 (reading may be changeable or erratic).

   Probable Cause: Sensor not connected, open sensor, open wiring between cabinet and sensor, wiring to sensor not shielded, or shield not properly grounded.

   Remedy: Check sensor and its wiring for continuity. Check that sensor cable is shielded and the shield grounded (see Connections under the Installation Section). If sensor has been deliberately disconnected for test purposes, see that the sensor terminals in the cabinet are temporarily connected together with a jumper.

5. TROUBLE: In Step 4 of Checking and adjusting Zero, the meter reads 2.0 V but in Step 5 the speed indicator reads at OFF (below zero rpm), or the pointer is against the stop above full scale.

   Probable Cause: Wiring between the cabinet and the speed indicator open, shorted or grounded.

   Remedy: Check and correct wiring to speed indicator.

6. TROUBLE: In Step 3 of Span Adjustment with the turbine at synchronous speed, the meter on the test panel reads correctly but the recorder reading is incorrect.

   Probable Cause: Recorder may not be in the speed recording mode but rather may be recording eccentricity or valve position.

   Remedy: See note at the beginning of the section of Preliminary Test and Adjustments.

7. TROUBLE: In Step 3 of Span Adjustment with the turbine at synchronous speed, the recorder reads zero rpm and the meter on the test panel reads 2.0 V.

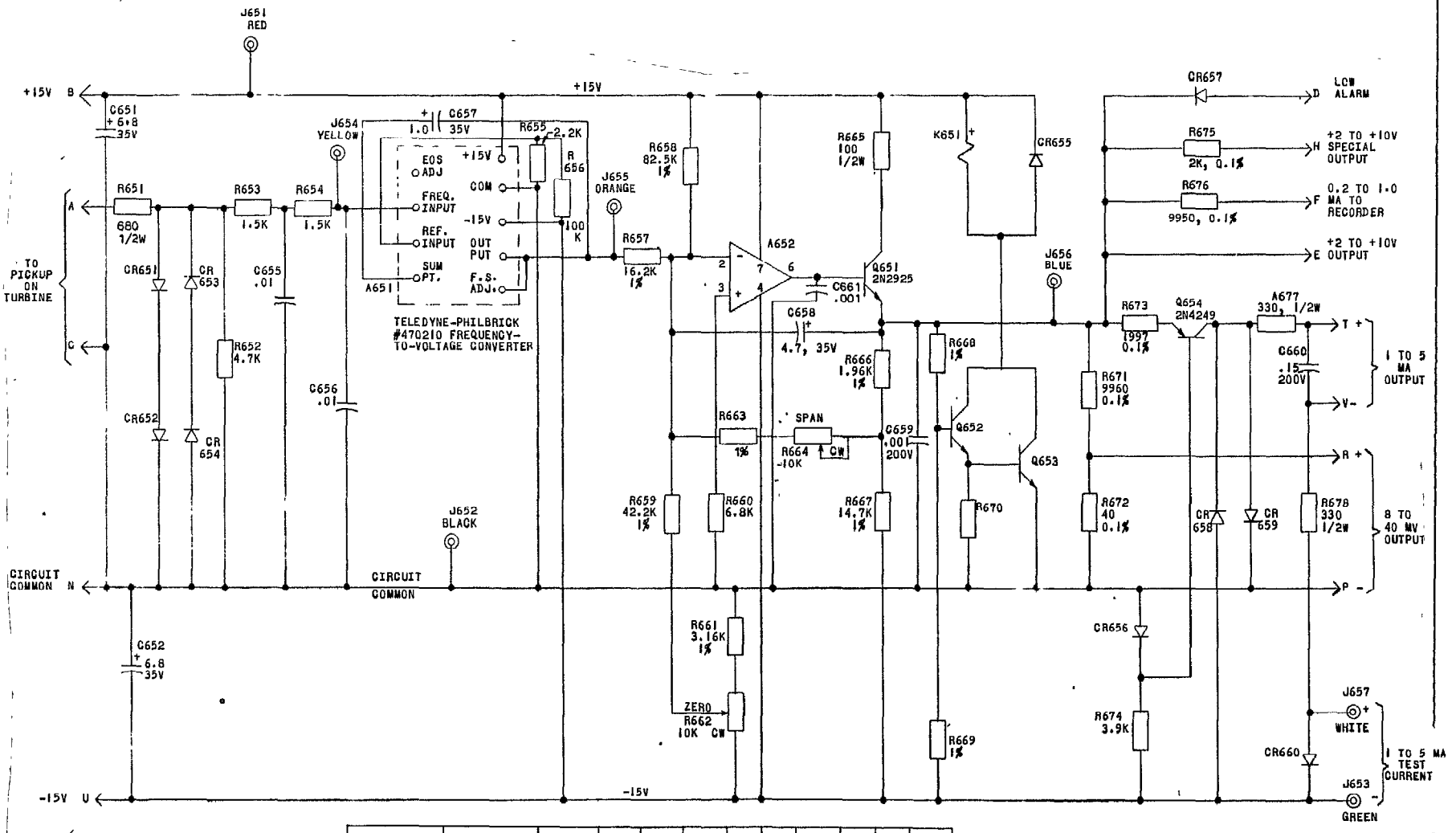
Probable Cause: No sensor voltage reaching the cabinet.

Remedy: Check with an ungrounded oscilloscope or electronic AC voltmeter to verify whether AC sensor voltage is reaching the cabinet terminals. Check the sensor and its associated pair of wires for open circuit, short circuit and grounds. Check for an abnormally large air gap between the sensor and its associated toothed wheel.

REPAIR AND REPLACEMENT PARTS

Should a circuit board be damaged or become inoperative, it is recommended that the board be replaced. A replacement circuit board should be ordered by its Catalog No. (see Specifications). Also give the model number of the turbine supervisory instrument system. Circuit boards, may if necessary, be returned to the factory for repairs. To order a replacement circuit board or for information on repair service, contact the nearest Installation and Service Engineering Office of General Electric Company.

NOTES:
1. RESISTANCE VALUES ARE IN OHMS, AND CAPACITANCE VALUES IN MICROFARADS. K=1000.
2. ALL DIODES ARE TYPE 1N645
3. RESISTOR TOLERANCE IS ± 5%, AND WATTAGE 1/4W, UNLESS OTHERWISE NOTED.
4. CAPACITORS ARE 100V, UNLESS OTHERWISE NOTED.
5. A652 FAIRCHILD U5B7741393 (UA741C) OR EQUAL.
6. KEY LOCATION S.



CIRCUIT BOARD MODEL NO.	ECCENTRICITY CUTOFF RELAY	K651	Q652	Q653	R668	R669	R670	CR655	C653	C654	R663
357700PB100A1	YES	INCLUDED	2N2925	2N2925	17.8K	75K	10K	IN645	.001	.001	10K
357700PB100B1	NO	OMIT	OMIT	OMIT	OMIT	OMIT	OMIT	OMIT	OMIT	OMIT	10K
357700PB100C1	NO	OMIT	OMIT	OMIT	OMIT	OMIT	OMIT	OMIT	OMIT	OMIT	16.2K

	TURBINE SYNCH. SPEED	VOLTAGE AT J655	
		AT SYNCH. SPEED	FULL SCALE
A1,B1	1800 OR 3600 RPM	-4.75V	-6.6V
A1,B1	1500 OR 3000 RPM	-4.75V	-7.9V
C1	3600 RPM	-3.56V	-4.95V

Figure 1
Elementary Diagram for Speed Amplifier Board, Frequency-Actuated Type
For Use with Proximity Pickup and Toothed Wheel (44C307501)

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